A Radar Vegetation Index for Crop Monitoring Using Compact Polarimetric SAR Data

Dipankar Mandal, *Student Member, IEEE*, Debanshu Ratha, *Student Member, IEEE*, Avik Bhattacharya, *Senior Member, IEEE*, Vineet Kumar, *Member, IEEE*, Heather McNairn, Yalamanchili S. Rao, *Member, IEEE*, and Alejandro C. Frery, *Senior Member, IEEE*

***Abstract*— Crop growth monitoring using compact-pol synthetic aperture radar (CP-SAR) data is gaining attention with the rapid advancements toward operational applications. In this article, we propose a vegetation index for compact polarimetric (CP) SAR data [compact-pol radar vegetation index (CpRVI)]. The CpRVI is derived using the concept of a geodesic distance between the Kennaugh matrices projected on a unit sphere. This distance is utilized to compute a similarity measure between the observed Kennaugh matrix and the Kennaugh matrix of an ideal depolarizer (a realization of vegetation canopy). The similarity measure is then modulated with a scaled quantity derived from the scattering power ratio of the same and opposite sense polarization with respect to the transmitted circular polarization. In this article, we utilize time-series-simulated RADARSAT Constellation Mission (RCM) compact-pol SAR data (RH-RV) obtained from the full-pol RADARSAT-2 observations during the soil moisture active passive (SMAP) validation experiment 2016 (SMAPVEX16-MB) campaign in Manitoba, Canada, to assess the proposed vegetation index. Among the various crops grown in this region, in particular, we analyze the growth stages of wheat and soybean due to their different canopy structures. A temporal analysis of the proposed CpRVI with crop biophysical parameters [the plant area index (PAI) and vegetation water content (VWC)] at different phenological stages confirms the trend of CpRVI with the plant growth. Nevertheless, variations of CpRVI values are apparent with different plant densities for both the crop types. Also, the linear regression analysis confirms that the CpRVI values significantly correlate with PAI ($r = 0.72$ and 0.85) and VWC ($r = 0.62$ and 0.75) for both wheat and soybean. We observed good retrieval of PAI and VWC for both wheat and soybean.**

***Index Terms*— Compact-pol, crop, geodesic distance (GD), RADARSAT Constellation Mission (RCM), vegetation index.**

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Dipankar Mandal, Debanshu Ratha, Avik Bhattacharya, Vineet Kumar, and Yalamanchili S. Rao are with the Microwave Remote Sensing Laboratory, Centre of Studies in Resources Engineering, IIT Bombay, Mumbai 400076, India (e-mail: dipankar.agrilengg@gmail.com; debanshu.ratha@iitb.ac.in; avikb@csre.iitb.ac.in; vineetk008@iitb.ac.in; ysrao@csre.iitb.ac.in).

Heather McNairn is with the Agriculture and Agri-Food Canada, Ottawa, ON K1A0C6, Canada (e-mail: heather.mcnairn@canada.ca).

Alejandro C. Frery is with the Laboratório de Computação Científica e Análise Numérica (LaCCAN), Universidade Federal de Alagoas, Maceió 57072-900, Brazil, and with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: acfrery@laccan.ufal.br).

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I. Introduction

POLARIMETRIC SAR data have provided exceptional opportunities to track temporal changes in vegetation conditions and to map crop characteristics over large spatial extents [1], [2]. With the advent of full-pol SAR sensors, it is now possible to monitor crop growth by relating the physical scattering mechanisms extracted from the data with crop phenology [3]. Several studies have reported the use of backscatter intensity ratios and polarimetric decomposition parameters as a proxy for crop growth [4]–[7].

Similar to the spectral indices that are well established in optical remote sensing, a vegetation index derived from SAR data could be an alternative for crop growth monitoring. In this direction, Kim and van Zyl [8] made the first attempt by introducing the radar vegetation index (RVI) using a measure of scattering randomness from vegetation targets. The RVI was formulated by modeling the vegetation canopy as a collection of randomly oriented dipoles. Several studies have utilized the RVI for crop growth monitoring and biophysical parameter estimation [9]–[11].

Even though full polarimetric SAR sensors offer unique performances in terms of the retrieval of rich quantitative information, the advantages of compact polarimetric (CP) SAR [12] systems for several Earth and planetary applications are gaining attention [13]–[17]. Compact-pol synthetic aperture radar (CP-SAR) modes have advantages over full-pol measurements in terms of larger swath width and less data volume at the expense of less polarimetric information.

In the literature, the utility of CP-SAR data is explored by comparing its performance with dual- and full-pol SAR data for the analysis of different land features [16], [18]–[25]. In particular, the potential of CP-SAR data has also been investigated for agricultural applications, such as crop growth monitoring [26]–[31], biophysical parameter retrieval [32], [33], and soil moisture retrieval [34]–[37], using real and simulated CP-SAR data from full-pol observations.

The potential of CP-SAR modes might hold significant interest from an operational perspective for agricultural applications with planned time-series data sets. These could be globally obtained from multiple SAR satellites considering the rapid expansion of constellations of satellites, such as the Canadian RADARSAT Constellation Mission (RCM),

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SAOCOM (TOPSAR with experimental CP-mode), and the NASA-ISRO SAR (NISAR) L- and S-band mission.

In a recent study, McNairn *et al.* [38] showed that the simulated CP parameters correlate with the normalized difference vegetation index (NDVI) derived from optical sensors. Homanouyi *et al.* [39] assessed the volume to odd-bounce scattering power ratio (P_v/P_s) derived from a scattering power decomposition method from the simulated CP-SAR data for vegetation condition monitoring. The P_v/P_s ratio shows a good correlation with NDVI and crop biomass for several crop types. However, as the crop canopy attained its peak growth during the end of the vegetative stage and fruit development, the magnitude of P_v/P_s was overestimated relative to that obtained from full-pol data. Nevertheless, these studies confirm that a derived CP-SAR metric similar to NDVI is essential for agricultural applications. Even though the RVI is a good alternative, it is restricted to the use of full-polarimetric SAR data. Thus, there is a need for an RVI for CP-SAR data. It is noteworthy that CP-SAR mode will become an important mode for the operational monitoring of Canada's agricultural extent.

In this article, we have proposed a new vegetation index, namely, Compact-pol RVI (CpRVI), for CP-SAR data using the concept of a geodesic distance (GD) between two Kennaugh matrices projected on unit sphere [40], [41]. In recent work, the authors have introduced a novel vegetation index [42] for full-pol SAR data using the generalized volume scattering model (GVSM) proposed by Antropov *et al.* [43]. This model is a function of two parameters that correspond to the copolarized ratio and cross-polarized correlation coefficient obtained from full-pol SAR data. However, for CP-SAR data, we have utilized the ideal depolarizer (ID) as a volume model that converts all Stokes vectors into a randomly polarized wave [44]. We compute a similarity measure between the observed Kennaugh matrix and a Kennaugh matrix corresponding to the ID. The formulation of CpRVI is direct and intuitive, as well as easy to implement. We tested the CpRVI with simulated RCM CP-SAR mode data from the full-pol RADARSAT-2 observations over the Joint Experiment for Crop Assessment and Monitoring (JECAM) test site in Carman (Manitoba), Canada. Also, we perform comparative analysis for the full crop season between CpRVI and full-pol vegetation indices.

The rest of the manuscript is organized in the following order. Section II describes the formulation of the Kennaugh matrix for CP-SAR data, the similarity measure using GD, and the formulation of CpRVI. Section III briefly describes the study area and data sets. Section IV discusses the results in detail by relating the CpRVI with two biophysical parameters: the plant area index (PAI) and vegetation water content (VWC) for wheat and soybean. Finally, the work is succinctly summarized and concluded in Section V.

II. Methodology

A. Kennaugh Matrix

In PolSAR, the 2×2 complex scattering matrix $\mathbf{S}$ encompasses complete polarimetric information about backscattering from targets

$$\mathbf{S} = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{1}$$

where the subscripts H and V denote the horizontal and vertical polarizations, respectively. Following the reciprocity theorem in the case of a monostatic radar, $S_{\mathrm{HV}} = S_{\mathrm{VH}}$.

A suitable representation of PolSAR data in terms of power is given by the 4×4 real Kennaugh matrix $\mathbf{K}$ defined as

$$\mathbf{K} = \frac{1}{2}\mathbf{A}^{*}\left(\mathbf{S} \otimes \mathbf{S}^{*}\right)\mathbf{A}^{-1}, \quad \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & j & -j & 0 \end{bmatrix} \tag{2}$$

where $\otimes$ is the Kronecker product, and $j = \sqrt{-1}$. The Kennaugh matrix for the incoherent case can also be obtained from the elements of the coherency matrix $\mathbf{T}$ in the following manner in (3), as shown at the bottom of this page, where T_{ij} is the (i, j)th entry of $\mathbf{T}$, and $\Re$ and $\Im$ denote the real and imaginary parts of a complex number.

Thus, the coherent elementary targets represented by $\mathbf{S}$ and the distributed targets usually represented by $\mathbf{T}$ (or the covariance matrix $\mathbf{C}$) can be suitably transformed into their $\mathbf{K}$ matrix representation using (2) and (3), respectively. Table I presents the Kennaugh matrix forms of some well-known elementary scatterers and the ID from PolSAR literature. These include symmetric scatterers such as trihedral (t), cylinder (c), dipole (dp), quarter waves ($\pm 1/4$), narrow dihedral (nd), dihedral (d), asymmetric scatterers, such as left and right helices (lh and rh), and ID.

It is noteworthy that different scatterers pertain to the description of features of interest in a PolSAR image. Thus, using a suitable distance measure to compare observations with the scattering models provides a good starting point for several applications. From this perspective, GD to compare two Kennaugh matrices was first proposed in [40] and,

$$\mathbf{K} = \begin{bmatrix} \dfrac{T_{11}+T_{22}+T_{33}}{2} & \Re(T_{12}) & \Re(T_{13}) & \Im(T_{23}) \\ \Re(T_{12}) & \dfrac{T_{11}+T_{22}-T_{33}}{2} & \Re(T_{23}) & \Im(T_{13}) \\ \Re(T_{13}) & \Re(T_{23}) & \dfrac{T_{11}-T_{22}+T_{33}}{2} & -\Im(T_{12}) \\ \Im(T_{23}) & \Im(T_{13}) & -\Im(T_{12}) & \dfrac{-T_{11}+T_{22}+T_{33}}{2} \end{bmatrix} \tag{3}$$

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TABLE I
KENNAUGH MATRICES FOR ELEMENTARY TARGETS AND THE ID

Target	Row 1	Row 2	Row 3	Row 4
$\mathbf{K}_t$	$1\ 0\ 0\ 0$	$0\ 1\ 0\ 0$	$0\ 0\ 1\ 0$	$0\ 0\ 0\ -1$
$\mathbf{K}_c$	$5/8\ \ 3/8\ \ 0\ \ 0$	$3/8\ \ 5/8\ \ 0\ \ 0$	$0\ \ 0\ \ 1/2\ \ 0$	$0\ \ 0\ \ 0\ \ -1/2$
$\mathbf{K}_{dp}$	$1\ -1\ 0\ 0$	$-1\ 1\ 0\ 0$	$0\ 0\ 0\ 0$	$0\ 0\ 0\ 0$
$\mathbf{K}_{+1/4}$	$1\ 0\ 0\ 0$	$0\ 1\ 0\ 0$	$0\ 0\ 0\ 1$	$0\ 0\ 1\ 0$
$\mathbf{K}_{-1/4}$	$1\ 0\ 0\ 0$	$0\ 1\ 0\ 0$	$0\ 0\ 0\ -1$	$0\ 0\ -1\ 0$
$\mathbf{K}_{nd}$	$5/8\ \ 3/8\ \ 0\ \ 0$	$3/8\ \ 5/8\ \ 0\ \ 0$	$0\ \ 0\ \ -1/2\ \ 0$	$0\ \ 0\ \ 0\ \ 1/2$
$\mathbf{K}_d$	$1\ 0\ 0\ 0$	$0\ 1\ 0\ 0$	$0\ 0\ -1\ 0$	$0\ 0\ 0\ 1$
$\mathbf{K}_{lh}$	$1\ 0\ 0\ -1$	$0\ 0\ 0\ 0$	$0\ 0\ 0\ 0$	$-1\ 0\ 0\ 1$
$\mathbf{K}_{rh}$	$1\ 0\ 0\ 1$	$0\ 0\ 0\ 0$	$0\ 0\ 0\ 0$	$1\ 0\ 0\ 1$
$\mathbf{K}_{\mathrm{ID}}$	$1\ 0\ 0\ 0$	$0\ 0\ 0\ 0$	$0\ 0\ 0\ 0$	$0\ 0\ 0\ 0$

since then, is used as an adaptable measure for several applications [40]–[42], [45]–[47].

B. GD and Its Applications

GD is motivated by the task of measuring the distance over a manifold (i.e., higher dimensional surface) intrinsically without leaving the surface along the shortest path (or geodesic). For example, the unit sphere is one of the simplest manifolds that are often encountered in several applications.

By definition, a great circle is the circle of radius equal to that of the sphere, and the shortest path between two points on the unit sphere is along the great circle passing through the two points. The distance measured along this circle is GD on the unit sphere. In general, this notion can be easily generalized to a higher dimensional real unit sphere.

In the context of PolSAR data, the space of 4×4 real matrices where all the Kennaugh matrices reside forms a convenient space to study GD on the unit sphere centered at the origin. Ratha *et al.* [40] proposed GD for Kennaugh matrices on this unit sphere. GD between two arbitrary Kennaugh matrices ($\mathbf{K}_1$ and $\mathbf{K}_2$) is defined as

$$\mathrm{GD}(\mathbf{K}_1, \mathbf{K}_2) = \frac{2}{\pi} \cos^{-1} \frac{\mathrm{Tr}\left(\mathbf{K}_1^T \mathbf{K}_2\right)}{\sqrt{\mathrm{Tr}\left(\mathbf{K}_1^T \mathbf{K}_1\right)} \sqrt{\mathrm{Tr}\left(\mathbf{K}_2^T \mathbf{K}_2\right)}} \tag{4}$$

where Tr is the trace operator, and the superscript T denotes transpose. The $2/\pi$ factor is used to normalize its range to $[0, 1]$. GD is adaptively utilized in several applications [40]–[42], [45]–[48] for full-polarimetric SAR data. The versatility of GD is due to its elegant properties, such as boundedness, scale invariance, and invariance under the orthogonal transformation of the polarization basis. A detailed treatment of GD with rigorous mathematical derivations of its useful properties and new roll invariant parameters using elementary models are detailed in [48].

Using specific scattering models in GD for particular applications, Ratha *et al.* [45] proposed a novel radar built-up index (RBUI) that utilizes GD for observations from even bounce scatterers (nd, d) and helices (lh, rh) for extracting built-up area from PolSAR data. However, the standard elementary models do not approximate well the scattering from vegetation. Hence, the concept of working with distributed targets in the form of volume scattering models for vegetation monitoring is prevalent in PolSAR literature. In this respect, Antropov *et al.* [43] proposed a generalized two-parameter volume scattering model to study backscattering from vegetation. Most of the volume models in the PolSAR literature follow this generalized volume model as a particular case by fixing its two parameters: γ and ρ. The parameters γ and ρ correspond to the copolarized channel ratio and the copolarized channel correlation coefficient, respectively.

Ratha *et al.* [42] synergistically combine GD of the observation with the generalized volume model, as well as GDs from even and odd bounce scatterers to propose an RVI [generalized RVI (GRVI)] for full-pol SAR data. The GRVI is compared with RVI proposed by Kim and Van Zyl [8] in terms of its correlation with biophysical parameters, such as the PAI and volumetric water content. We obtain GD-based indices for full polarimetric SAR data represented by the Kennaugh matrices without any assumption on the polarization mode, so they can also be utilized for compact-pol SAR data.

C. Compact-Pol RVI

The hybrid CP mode measures a projection of the 2×2 complex scattering matrix $\mathbf{S}$ as

$$\begin{bmatrix} E_{\mathrm{CH}} \\ E_{\mathrm{CV}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \begin{bmatrix} 1 \\ \pm i \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} \pm i S_{\mathrm{HV}} \\ S_{\mathrm{VH}} \pm i S_{\mathrm{VV}} \end{bmatrix} \tag{5}$$

where the subscript C can be either the left-hand circular (LHC) transmit with a $+$ sign or the right-hand circular (RHC) transmit with a $-$ sign. The 2×2 covariance matrix is then obtained from the elements of the scattering vector as

$$\langle [\mathbf{C}] \rangle = \begin{bmatrix} \langle |E_{\mathrm{CH}}|^2 \rangle & \langle E_{\mathrm{CH}} E_{\mathrm{CV}}^* \rangle \\ \langle E_{\mathrm{CV}} E_{\mathrm{CH}}^* \rangle & \langle |E_{\mathrm{CV}}|^2 \rangle \end{bmatrix}. \tag{6}$$

For CP-SAR data, the 4×1 Stokes vector $\vec{\mathbf{g}}$ can be written in terms of the elements of the 2×2 covariance matrix $\mathbf{C}$ as

$$\vec{\mathbf{g}} = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} C_{11} + C_{22} \\ C_{11} - C_{22} \\ C_{12} + C_{21} \\ \pm j(C_{12} - C_{21}) \end{bmatrix} \tag{7}$$

where $\pm$ corresponds to left and right circular polarization respectively. The Kennaugh matrix ($\mathbf{K}_{\mathrm{CP}}$) for CP-data can be written in terms of the elements of the Stokes vector as

$$\mathbf{K}_{\mathrm{CP}} = \begin{bmatrix} g_0 & 0 & g_2/2 & 0 \\ 0 & 0 & 0 & g_1 \\ g_2/2 & 0 & 0 & 0 \\ 0 & g_1 & 0 & g_3/2 \end{bmatrix}. \tag{8}$$

In the absence of any volume scattering models to suitably describe scattering from vegetation in CP-SAR data, ID is utilized to measure randomness from distributed targets. Moreover, GDs from the odd (i.e., trihedral) and even bounce (i.e., dihedral) scatterers that appear in GRVI [42] are suitably replaced by the received echo powers from the same circular (SC) and the opposite circular (OC) sense to the transmitted polarization.

In this light, we propose a novel CpRVI (9) by utilizing the measure of similarity $f_{\mathrm{ID}} = 1 - (3/2)\mathrm{GD}_{\mathrm{ID}}$ between a

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Kennaugh matrix for an ID $\mathbf{K}_{\mathrm{ID}}$ and the observed Kennaugh matrix $\mathbf{K}$ using GD: $\mathrm{GD}_{\mathrm{ID}} = \mathrm{GD}(\mathbf{K}, \mathbf{K}_{\mathrm{ID}})$.

The quantity $2/3$ is the maximum GD between an ID ($\mathbf{K}_{\mathrm{ID}}$) and an observed Kennaugh matrix $\mathbf{K}$. Moreover, the similarity measure f_{ID} is zero for any other elementary target, and therefore, $\mathrm{CpRVI} = 0$.

The parameter β in (10) is proportional to the ratio of the minimum to the maximum backscattering power in the same (SC) and the opposite (OC) sense to the transmitted circular polarization

$$\mathrm{CpRVI} = \beta\, f_{\mathrm{ID}}, \quad 0 \le \mathrm{CpRVI} \le 1 \tag{9}$$

$$\beta = \left(\frac{p}{q}\right)^{2\,(3/2\,\mathrm{GD}_{\mathrm{ID}})} \tag{10}$$

$$p = \min\{\mathrm{SC}, \mathrm{OC}\}, \quad q = \max\{\mathrm{SC}, \mathrm{OC}\} \tag{11}$$

$$\mathrm{SC} = \frac{g_0 - g_3}{2}, \quad \mathrm{OC} = \frac{g_0 + g_3}{2} \tag{12}$$

where g_0 (the first element of the Stokes vector) is a measure of the total average received power and g_3 (the fourth element of the Stokes vector) is a measure of the average received power in circular polarization.

The proportion of the power that is received by the radar in opposite-sense circular polarization to that transmitted is $(g_0 + g_3)/2$. For example, $\mathrm{OC} = (g_0 + g_3)/2 = |E_R|^2$ for left-circular (L) polarization on transmit, where $|E_R|^2$ is the intensity of the right-circular component. Such a change of polarity occurs when an electromagnetic (EM) wave undergoes an odd number of reflections from a target.

Similarly, $(g_0 - g_3)/2$ is a measure of the power received by the radar in the same-sense circular polarization as it was transmitted, which represents an even number of reflections from a target. For example, $\mathrm{SC} = (g_0 - g_3)/2 = |E_L|^2$ for left-circular (L) polarization on transmit, where $|E_L|^2$ is the intensity of the left-circular component.

The role of β is similar to that of the damping factor used in GRVI. It ensures the imbalance between the SC and OC returns, which is identical to the case for even and odd bounce returns for GRVI. For purer targets with either an odd or an even bounce-dominant scattering mechanism, there will be a substantial imbalance between the SC and OC returns. This imbalance leads to a smaller value of β, thereby resulting in a lower value of CpRVI. While for random scattering media (namely, vegetation), which do not have an overall preference for SC or OC returns, the value of β is high (i.e., close to 1), which leads to a higher value of CpRVI.

Theoretically, for a smooth bare surface, the similarity measure, $f_{\mathrm{ID}} \approx 0$, between the observed Kennaugh matrix $\mathbf{K}$ and $\mathbf{K}_{\mathrm{ID}}$, hence, for this case, $\beta \ll 1$, as $\mathrm{SC} \ll \mathrm{OC}$. This scenario also indicates a lower circular polarization ratio (CPR) with $\mathrm{CpRVI} \approx 0$. In the case of a random volume scattering (with little to no polarization structure), $f_{\mathrm{ID}} \approx 1$ and $\beta \approx 1$. The SC and OC components would then be similar (equal to $g_0/2$) for such random volume cases with $\mathrm{CpRVI} \approx 1$. Fig. 1 shows the variation of f_{ID} versus β with CpRVI.

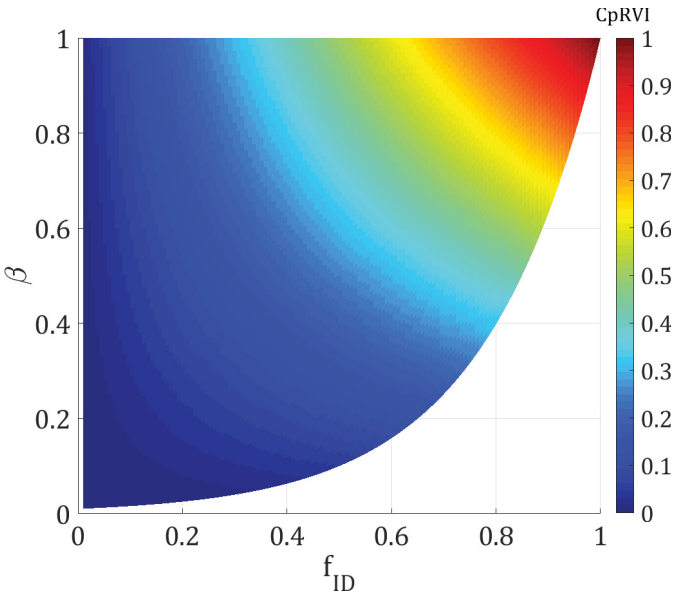


Fig. 1. Dependence of CpRVI on f_{ID} and β.

III. Study Area and Data Set

This article was conducted over the JECAM test site in Carman, Manitoba (Canada), shown in Fig. 2. The site covers an area of 26 km × 48 km and is characterized by various crops and soil conditions. The major crops grown in this test site are wheat, soybean, canola, and corn. According to the 2016 Province Report [49], the seeding of annual crops in Manitoba started at the end of April to mid-May with harvesting commencing in August. The *in situ* measurements were collected over the area with near coincident satellite passes as a part of the Soil Moisture Active Passive (SMAP) Validation Experiment 2016 (SMAPVEX16-MB) campaign [50].

A. In Situ Sampling Strategy

During the campaign, *in situ* measurements of crops and soil were collected in two separate periods (8 June to 22 June, and 9 July to 22 July 2016) over 50 agricultural fields. The nominal size of each field is around 800 m × 800 m. In each field, soil moisture was measured at sixteen sampling points, arranged in two parallel transects along the row direction (see Fig. 2). Sampling points were separated by approximately 75 m, and each transect was separated by 200 m. Soil moisture was measured at each point using hydra probes.

The vegetation was sampled in each field at three points (i.e., points 2, 11, and 14 in the first week, and 3, 10, and 13 in the second week). Vegetation sampling included measurement of PAI, wet biomass, plant height, density, and phenology through destructive and nondestructive methods. The SMAPVEX16-MB report [50] and [51] provides detailed descriptions of vegetation and soil sampling strategies.

B. SAR Data and Preprocessing

Several RADARSAT-2 images were acquired during the campaign in Fine Wide quad-pol mode (FQW); see Table II

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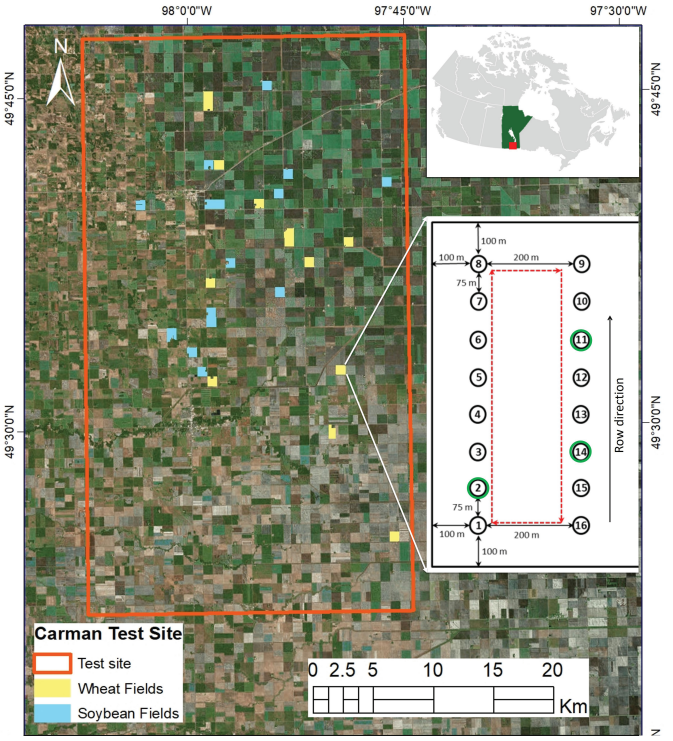


Fig. 2. Study area and sampling locations over the JECAM-Carman (Canada) test site. The layout of the 16 sampling locations within each field is highlighted. In each field, three points (i.e., points {2, 11, 14} or {3, 10, 13}) were selected for vegetation sampling.

for details of the satellite data used in this article. The selection of acquisition dates was based on *in situ* measurement periods.

Several polarization combinations at varying spatial resolutions and noise floors (ranging from −25 to −17 dB) would be available for RCM [52] that includes single polarization (i.e., HH, VV, HV, and VH), dual-pol (i.e., HH-VV, HH-HV, and VV-VH), full-pol (i.e., HH-HV-VH-VV), and compact-pol (i.e., RH-RV) modes.

In this article, we simulated the compact-pol (RH-RV) data from full-pol RADARSAT-2 data using the RCM Compact Polarimetry Simulator developed at the Canada Centre for Remote Sensing (CCRS) [16], [53]. The simulator produces the simulated SAR products based on the prelaunch specifications of RCM provided by the Canadian Space Agency. Specifications such as the spatial resolution and noise equivalent sigma zero (NESZ) of each mode are considered during the simulation process. The RCM simulator uses full-pol RADARSAT-2 images as input to simulate RCM data products of any of the RCM modes.

The RCM simulator first converts the RADARSAT-2 16-bit complex products (S_{HH}, S_{HV}, S_{VH}, and S_{VV}) to 32-bit float complex by applying the calibration coefficients specified for a particular product of RCM. Random noise is added to the complex imagery. In this article, the compact-pol (RH-RV) data are simulated with −19-dB NESZ for the high-resolution (5-m pixel size) imaging mode of RCM [52]. It is noteworthy that CP-SAR mode will become an important mode for the operational monitoring of Canada's agricultural extent.

TABLE II
SPECIFICATION OF RADARSAT-2 DATA ACQUISITIONS USED IN THIS ARTICLE

Acquisition date	Beam Mode	Incidence Angle Range (deg.)	Orbit
30/05/2016	FQ7W	24.98 to 28.32	Ascending
15/06/2016	FQ7W	24.98 to 28.32	Descending
23/06/2016	FQ7W	24.98 to 28.32	Ascending
09/07/2016	FQ7W	24.98 to 28.32	Descending
17/07/2016	FQ7W	24.98 to 28.32	Ascending
02/08/2016	FQ7W	24.98 to 28.32	Descending
10/08/2016	FQ7W	24.98 to 28.32	Ascending

Research over Canadian JECAM test sites has confirmed that despite its higher NESZ, simulated RCM CP-SAR data can accurately classify crops [54], [55]. As the NESZ of RCM is higher than the RADARSAT-2 system, there could be a decreased sensitivity to low backscatter values. Hence, care should be taken when deriving the polarimetric parameters from simulated RCM products.

In the simulation process, the complex scattering vector in RH and RV polarizations are then calculated from the S_{HH}, S_{HV}, S_{VH}, and S_{VV} elements, as given in (5). Here, it is important to note that the ellipticity, $\chi = -45°$, and orientation angle $\psi = 0°$ are considered constant values in the RCM simulator to generate a perfect circular transmit condition. Based on the RCM product specifications, the RH and RV imagery are then downsampled using a 2-D fast Fourier transformation technique to obtain the spatial resolution defined for the specific beam mode. The elements of the scattering vector in RH and RV polarizations are then stored in a product file along with a metadata information file as an RCM simulator output for each acquisition date (see Table II).

These simulated outputs are utilized further to generate the CpRVI. First, the 2×2 covariance matrix is obtained from the elements of the scattering vector, as given in (6). During the $\langle[\mathbf{C}]\rangle$ generation, a 2×2 multilook is applied in the range and azimuth directions. The Stokes vectors are then used to calculate the Kennaugh matrix $\mathbf{K}$. Finally, CpRVI images are obtained from the derived $\mathbf{K}$ for the simulated CP-SAR data. To reduce the effect of speckle, the CpRVI is calculated over a 5×5 averaging window. The CpRVI images are coregistered using ground control points (GCPs) with an RMSE ≤ 0.33 m and georeferenced. The sampling points are overlayed on the CpRVI images, and CpRVI values for each site (point measurement) are extracted from the coregistered CpRVI images over wheat and soybean fields with 3×3 window.

Also, a comparative analysis is performed for the full crop season between CpRVI and full-pol vegetation indices. The full-pol indices, namely, GRVI, and RVI are obtained from the seven full-pol RADARSAT-2 images (see Table II). The full-polarimetric C-band RADARSAT-2 images are multilooked by 2×2 (in the range and azimuth) to generate the coherency matrix $\mathbf{T}$. The elements of the $\mathbf{T}$ matrix are then used to calculate the Kennaugh matrix $\mathbf{K}$, as discussed in Section II-A. The GRVI images are generated from the derived $\mathbf{K}$ for each

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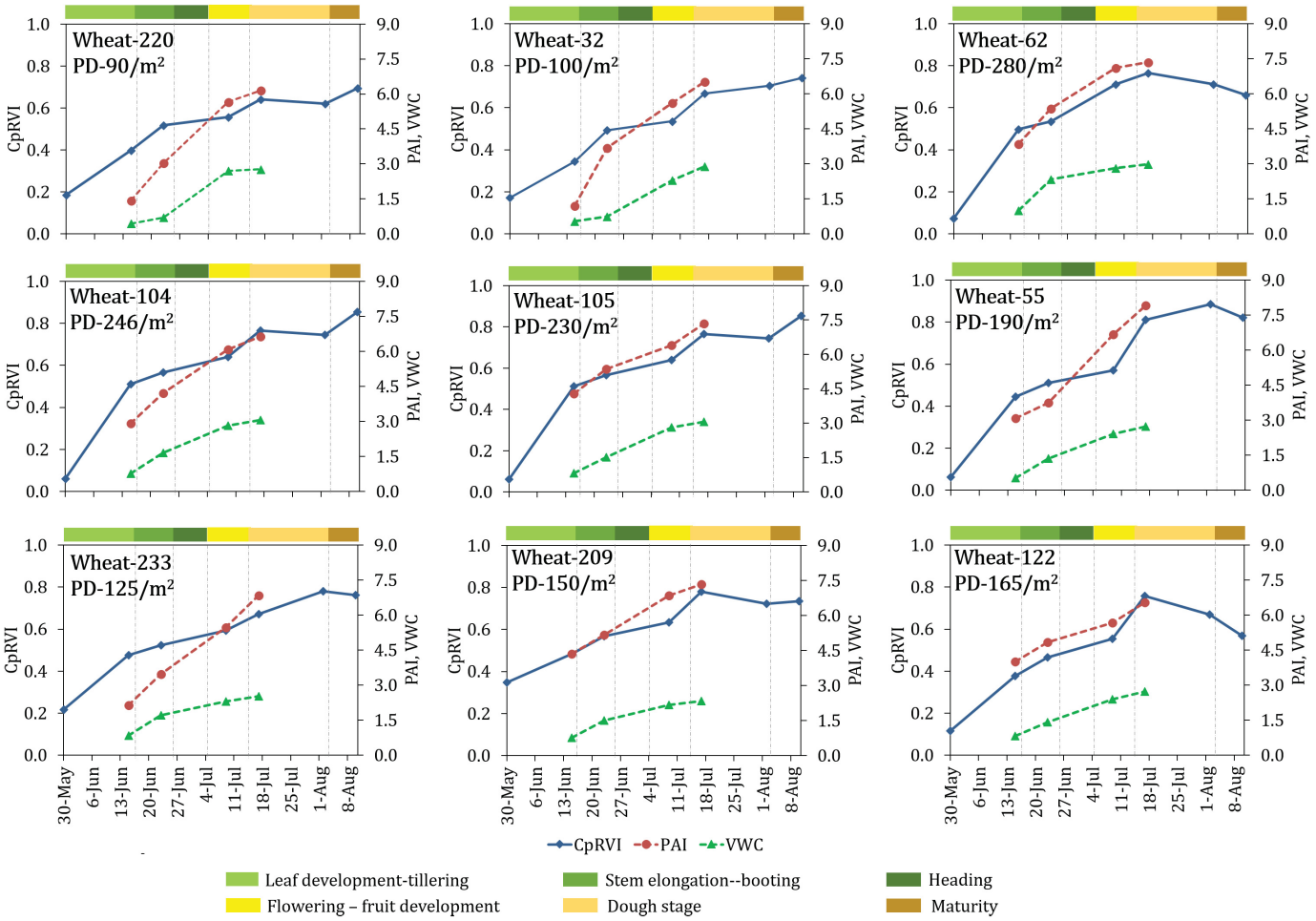


Fig. 3. Temporal pattern of CpRVI for several wheat fields at different crop growth stages. Wheat fields are selected based on variations in PD (number of plants per m^2 area). The *in situ* measurements of PAI ($m^2\,m^{-2}$) and VWC ($kg\,m^{-2}$) are plotted in secondary axis for each field.

acquisition over 5 × 5 moving overlapping window [42]. In a separate process flow, the RVI [8] images are obtained from the eigenvalue spectrum obtained from the **T** matrix [56]. Both the GRVI and RVI images are then coregistered using GCPs with an RMSE $\leq$ 0.54 m. The GRVI and RVI values for each sampling location (point measurements) are extracted over 3 × 3 moving overlapping window.

IV. Results and Discussion

The CpRVI for different sampling sites are generated from the simulated RCM data set, and we perform a seasonal analysis with different growth stages for wheat and soybean. Furthermore, the correlation of CpRVI with PAI ($m^2\,m^{-2}$) and VWC ($kg\,m^{-2}$) are analyzed for individual crops. Figs. 3 and 7 show the temporal trends of CpRVI averaged for three sampling points in each field. In total, 96 samples from 13 wheat fields and 43 samples from 14 soybean fields were used in the regression model (see Fig. 4 and Table IV). Consequently, the PAI and VWC are estimated using these models for both wheat and soybean crops.

A. Wheat (Triticum Aestivum)

Fig. 3 shows the temporal response of CpRVI over the phenological stages of wheat. For the analysis of CpRVI, field measurements are compared among three groups of which are separated based on plant density (PD: number of plants per square meter). The growth trends are similar irrespective of PD among several fields, i.e., an increase of PAI and VWC with crop development.

At the early stage of development (on 30 May), the magnitude of CpRVI is low ($\lessapprox$0.2) in the majority of the wheat plots. Alternatively, a high degree of polarization (DOP) values ($\approx 0.75 \pm 0.03$) suggests low roughness and randomness from the fields (see Fig. 5). The low CPR values ($\approx 0.25 \pm 0.02$) indicate high opposite sense (OC) received power, which is likely due to odd-bounce scattering phenomena from the bare soil with less vegetation cover at the early stage of leaf development [28], [31]. These low values of CpRVI indicate that the observed Kennaugh matrix would be similar to an elementary target (i.e., trihedral in this case).

Separation of CpRVI values among several fields becomes apparent when wheat has advanced from the leaf development to the tillering stage on 15 June. Fields with PD of $\approx$100 m^{-2} (Fields no. 220 and 32) have low CpRVI values ($\approx 0.37 \pm 0.05$) that are comparatively lower than wheat fields with high PD (125 and 190 m^{-2}). The *in situ* measurements of PAI and VWC are also comparatively higher ($>3.5\ m^2\,m^{-2}$ and $\approx 1.0\ kg\,m^{-2}$) for wheat fields with high PD.

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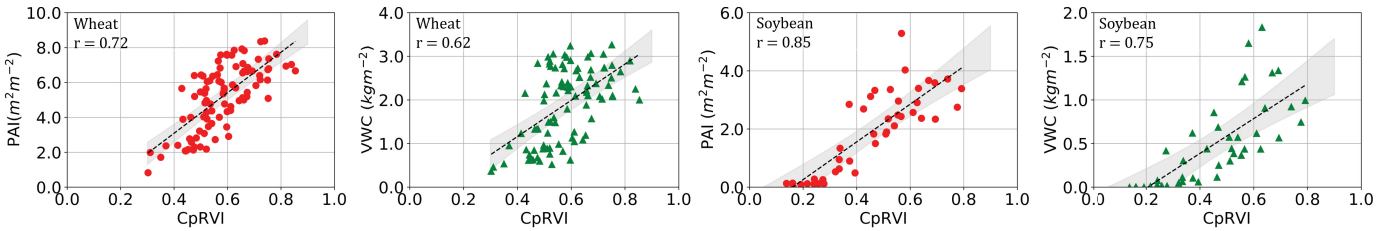


Fig. 4. Linear model correlations (r) between CpRVI with PAI ($m^2\,m^{-2}$) and VWC ($kg\,m^{-2}$) for wheat and soybean.

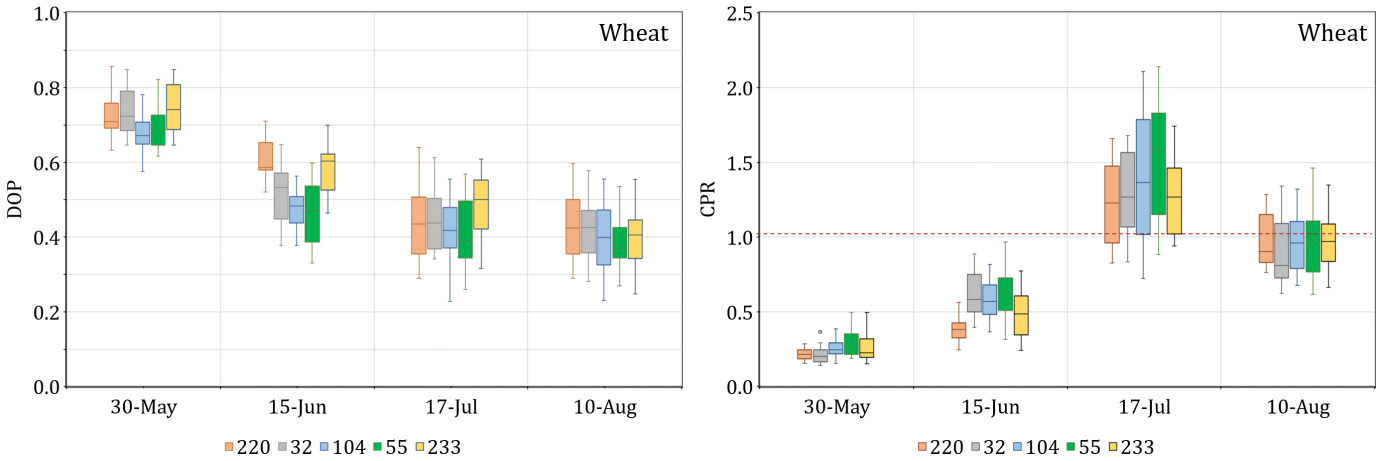


Fig. 5. Temporal variations of DOP and CPR for wheat fields (220, 32, 104, 55, and 233).

It is important to note that CpRVI values reached their maximum when crop advanced from stem elongation to early dough or fruit development stages on 17 July. CpRVI values reach up to 0.6 for low PD fields, while these values peak at ≈0.8 for high PD wheat fields. Such difference may be due to the high degree of randomness in scattering (DOP ≈ 0.4 on 17 July) from the canopy elements during the flowering to fruit development stages.

The *in situ* measurements of PAI and VWC at the flowering to fruit development stages confirm their increase up to approximately 6.5 $m^2\,m^{-2}$ and 3.0 $kg\,m^{-2}$, respectively. This behavior indicates the dominance of volume scattering in the fields, which might increase the similarity between the observed $\mathbf{K}$ and the ID $\mathbf{K}_{\mathrm{ID}}$. Also, the presence of even bounce scattering is often due to the interaction of the EM wave with the vertical stems of the wheat canopy and the underlying soil. This increases the SC component [31], thereby increasing the β factor in the CpRVI formulation. Moreover, CPR values >1.2 also confirm that the same sense (SC) power is higher than OC during this period (see Fig. 5).

The differential increase in CpRVI among the wheat fields is apparent in Fig. 3. Variations in the PD could cause dissimilarity with CpRVI in several fields, even if they are in the same phenological stage. Multiple interactions of the SAR signal cause high volume scattering in the dense canopy, which is observed for fields with high PD.

The rate of increase in CpRVI values slows down at the end of July after the stagnation of the vegetative growth and crops begin seed development. The ripening of wheat started during the first two weeks of August, followed by swathing and harvest in the last week of August. During this period, the DOP values become low (≈0.35), and CPR saturates to ≈1. Also, the SC and OC components are quite similar, which indicates high volume scattering. However, an ID does not exist in nature, and hence, as mentioned earlier, GD($\mathbf{K}$, $\mathbf{K}_{\mathrm{ID}}$) varies in the range $[0, 2/3]$. Also, the spatiotemporal changes in several wheat fields are apparent in Fig. 6.

The correlation analysis of CpRVI with PAI and VWC along with the confidence interval (CI) is shown in Fig. 4, which indicates the variation between these two biophysical parameters. The correlation coefficient (r) of CpRVI with PAI is 0.72 ($p = 0.005$, 95% CI: [0.60, 0.80]), which is slightly higher than VWC with $r = 0.62$ ($p = 0.006$, 95% CI: [0.48, 0.73]). Variations in CpRVI with PAI and VWC at later growth stages are likely due to scattering from the upper canopy layer (i.e., wheat heads). Similarly, Wu *et al.* [57] reported using a ground-based scatterometer experiment that the wheat heads dominate the total scattering power at the heading stage. However, during the ripening stage (when the heads become drier), the backscatter from the ground becomes dominant, and the backscatter power from the heads is insensitive to the moisture content. Also, variations in backscatter power are less prominent with changes in leaf area or biomass [58]. In fact, during the ripening stage of wheat, the backscatter power is insensitive to plant biomass [59] at low incidence angle (as used in the present case of RADARSAT-2 FQ7W mode of ≈25° incidence angle).

B. Soybean (Glycine Max)

Compared to wheat, the soybean canopy has more random structure [60] with secondary stems and randomly

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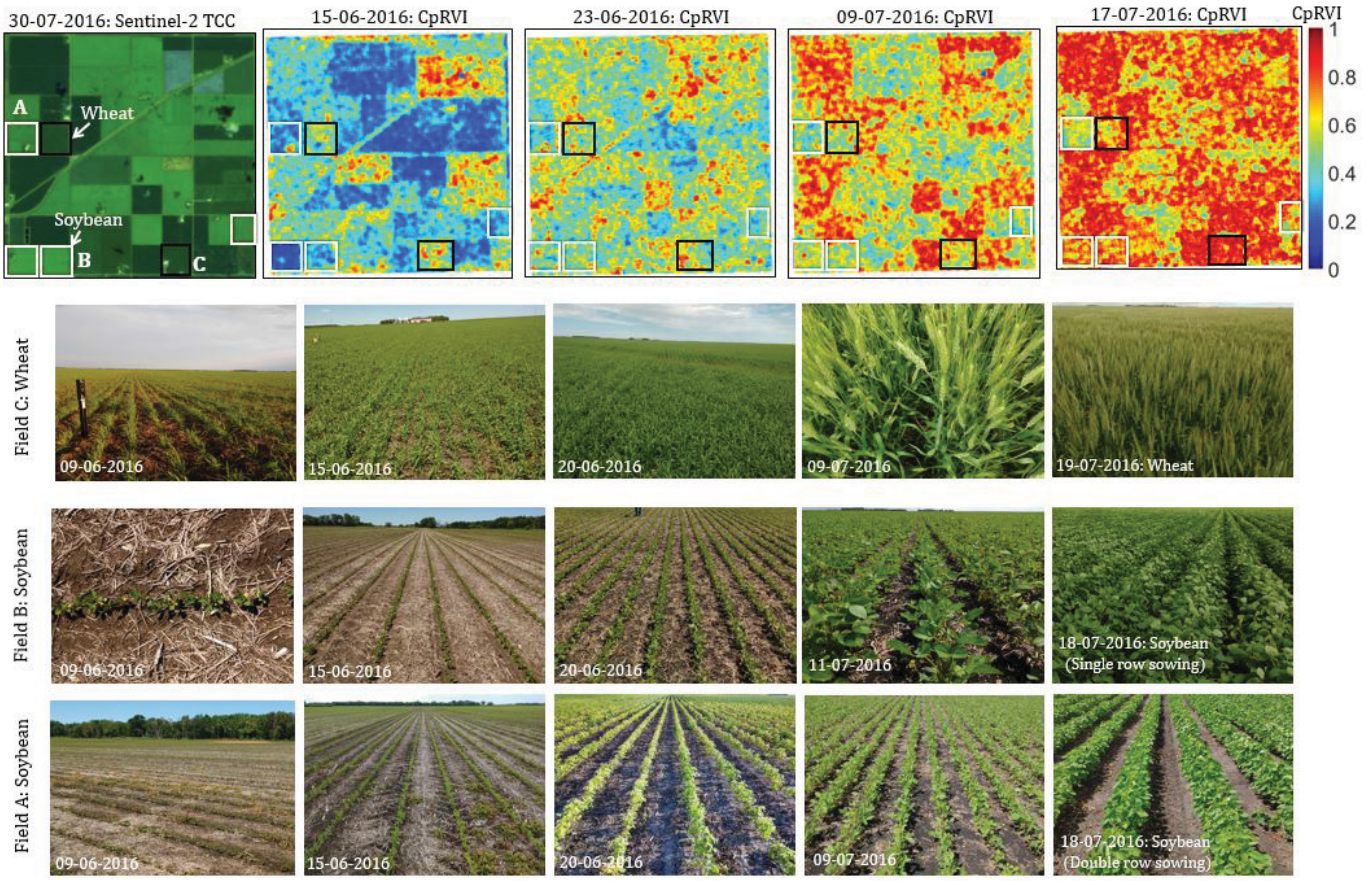


Fig. 6. Spatiotemporal variations of CpRVI within the wheat and soybean fields. The black and white polygons show the sampling fields of wheat and soybean, respectively. The field photographs during the campaign at specific instances are presented for wheat and soybean.

oriented leaves. Hence, unlike erectophiles (canopy elements have predominant vertical distribution), the response of CpRVI would be different for the random structure of the soybean canopy. The Manitoba weekly crop reports [49] suggest that soybean seeding was completed by the third week of May. Thus, crop development during the beginning of the SMAPVEX-16 campaign in June was primarily limited to vegetative growth. Subsequently, inflorescence emergence, flowering, and pod development started during the last week of July. The development of pods, ripening of seeds, and senescence are followed in August until the second week of September. Here, it is worth mentioning that the seeding pattern and the choice of row-plant spacing (single- and double-row plantings) varied among the fields (see Fig. 7).

Fig. 7 presents the temporal trends of CpRVI for several fields with different row spacing (RS) and plant count (PC) per meter length. The CpRVI values for each field increase rapidly as the vegetation growth increases from the early vegetative growth stages to the beginning of pod development. The CpRVI value is approximately 0.25 ± 0.06 at the end of the early leaf development stage (15 June). The high value of DOP (0.75 ± 0.05 in Fig. 8) also suggests less randomness from the soybean fields, which is primarily affected by the underlying soil as a result of less plant cover (PAI < 0.1 m^2 m^{-2}). The CPR value is approximately 0.3, which implies that the OC component is much higher than SC (see Fig. 8). Bare soil with shallow vegetation likely produces higher surface scattering. The dominance of surface scattering during the early stage was also reported in [39], where the volume to surface scattering power ratio in CP-SAR data was below 0.7.

With the increase in vegetation components, the variations in CpRVI values among several fields are apparent. The soybean fields are separated into three groups: group A (RS $<$ 40 cm and PC $<$ 40 per meter length of row) includes fields 72, 113, and 221; group B (RS $>$ 75 cm and PC $>$ 65 per meter length of row) includes fields 65, 82, and 228; and group C (RS $\approx$ 50 cm and PC $\approx$ 50 per meter length of row) includes fields 223, 231, and 232. During the pod development stage, the CpRVI values reach up to 0.75 ± 0.06 for group C.

Here, it is essential to note that despite having PC $>$ 65 m^{-1} for group B, the CpRVI values are comparatively lower (approximately 10%–15%) than fields of group C. The *in situ* measurements confirmed that fields in group B had wider RS (approximately 77 cm) and less canopy closure (see Fig. 6), while the number of plants along the row was higher (a double-row-seeded soybean). Therefore, the PAI of group B fields is lower than the fields of group C. In contrast to the fields in groups B and C, there are narrower RS (approximately 38 cm) and low PC (<40 per meter row) for fields in group A. Hence,

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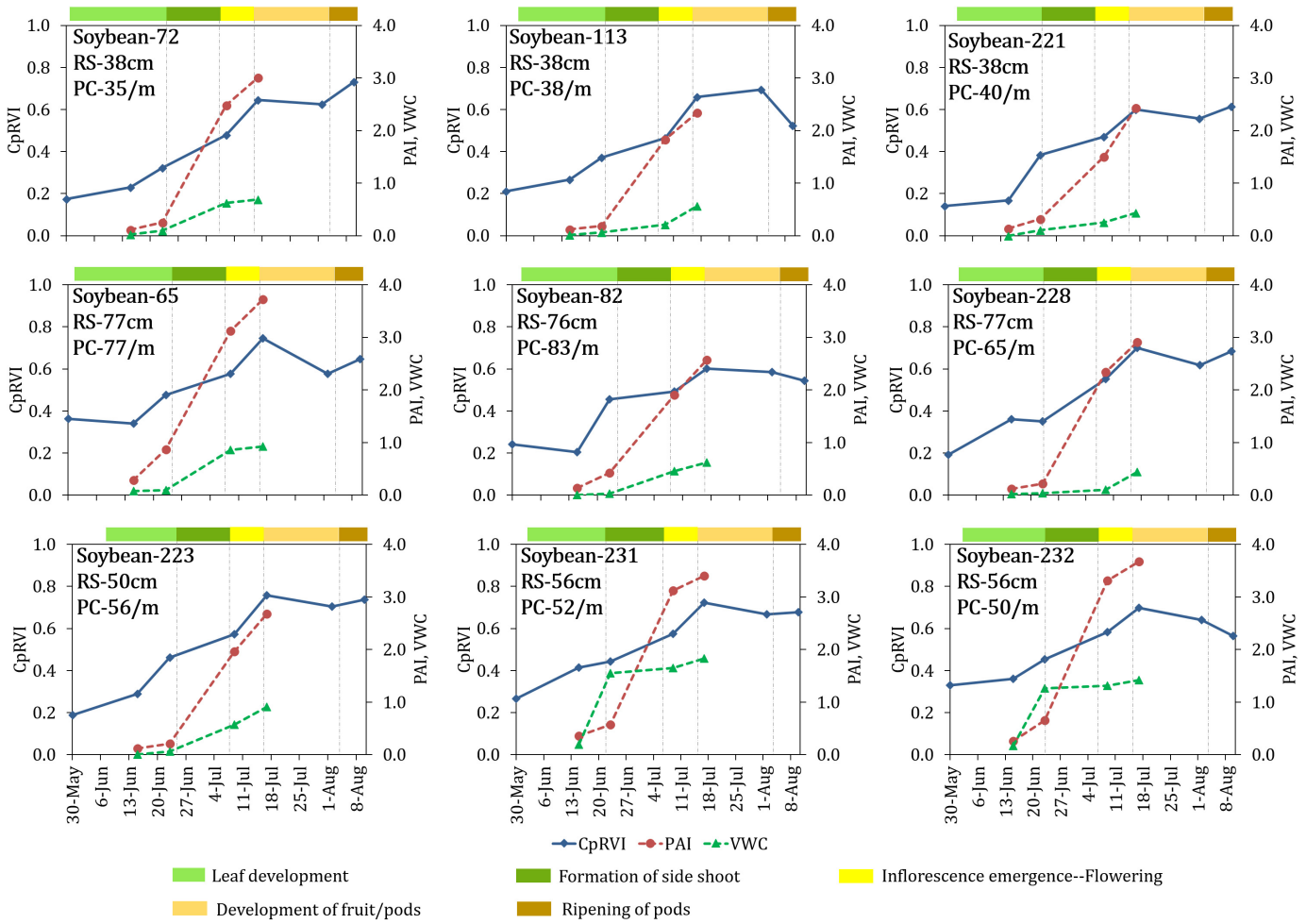


Fig. 7. Temporal pattern of CpRVI for several soybean fields at different crop growth stages. Soybean fields are selected based on variations in plant RS (cm) and PC (number of plants per meter of row length). The *in situ* measurements of PAI ($m^2\,m^{-2}$) and VWC ($kg\,m^{-2}$) are plotted in secondary axis for each field.

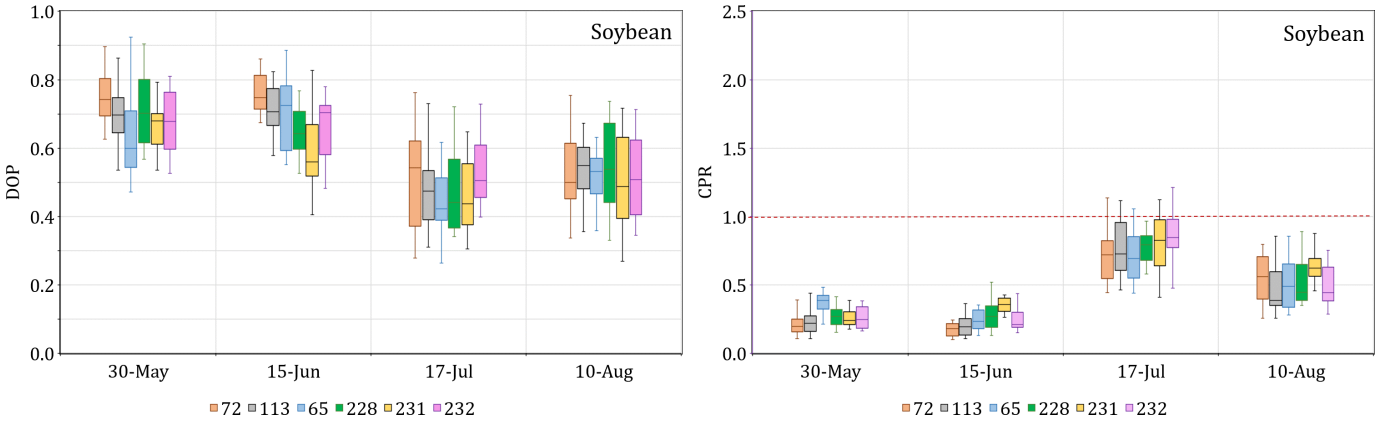


Fig. 8. Temporal variations of DOP and CPR for soybean fields (72, 113, 65, 228, 231, and 232).

more volume scattering is possible from the fields of group B, which, in turn, results in higher CpRVI. Wiseman *et al.* [61] also reported differences in volume scattering with such varied PD.

It is also apparent from Fig. 8 that low DOP and CPR values (approximately 0.85) during the pod development stage indicate an increase in the volume scattering component. In the case of group A fields, the CpRVI values are restricted within the range of 0.65 ± 0.05, which is comparatively lower than group C. Also, PAI and VWC are high during this stage.

The correlation plots shown in Fig. 4 indicate that the CpRVI values are correlated better with the PAI than VWC. It is observed that the correlation coefficients of the PAI and VWC with CpRVI are $r = 0.85$ ($p = 0.002$,

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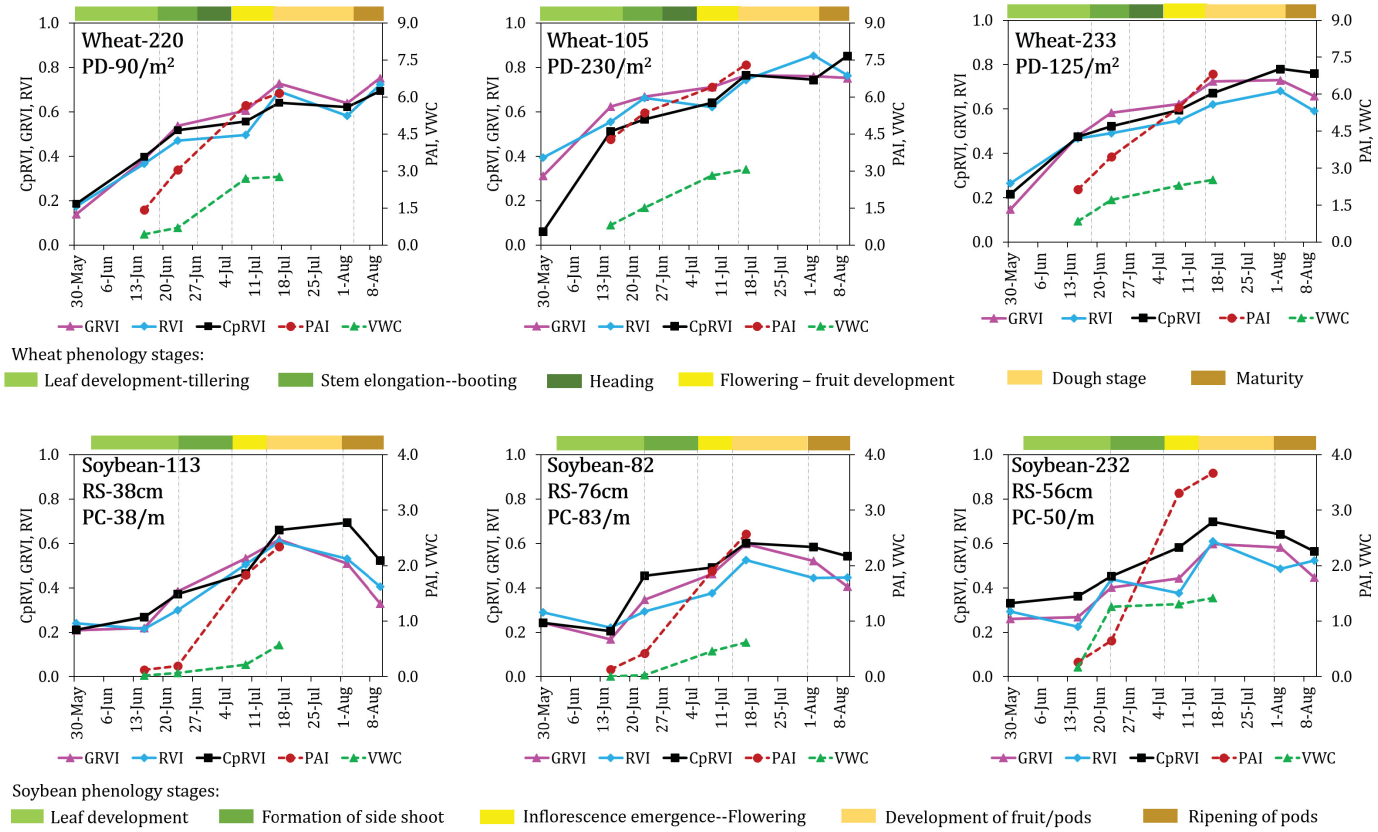


Fig. 9. Temporal pattern of vegetation indices (CpRVI, RVI, and GRVI) for wheat and soybean fields at different phenological stages. Wheat fields are selected based on variations in PD (number of plants per m^2). Soybean fields are selected based on variations in plant RS (cm) and PC (number of plants per meter of row length). The *in situ* measurements of PAI ($m^2\,m^{-2}$) and VWC ($kg\,m^{-2}$) are plotted in the secondary axis for each field.

95% CI: [0.75, 0.91]) and $r = 0.75$ ($p = 0.003$, 95% CI: [0.59, 0.85]), respectively. It is evident from Fig. 4 that for both PAI and VWC, the CpRVI values are low at the early vegetative stages (PAI $< 1.0\ m^2\,m^{-2}$ and VWC $> 0.35\ kg\,m^{-2}$). This phenomenon is likely because the CpRVI derived at the early growth stage is affected to a higher degree by the underlying soil rather than the vegetation canopy.

Fig. 6 shows the CpRVI maps for the four acquisition dates in over a subset of the test site. Both spatial and temporal variabilities are observed in these maps, which are similar to the spatiotemporal behavior of soybean, as shown in Fig. 7. During the second week of June, the soybean plants advanced from the unifoliate to the third trifoliate stage, which is well represented by the low values (<0.25) of CpRVI. The plant growth in subsequent weeks of June and the first week of July is apparent (increased to ≈0.55) in the CpRVI images for almost all soybean fields. The Manitoba weekly reports [49] inform the seventh trifoliate stage in the majority of the fields. Rapid growth is also evident in the subsequent weeks of July with soybean at its full vegetative growth. Following vegetative growth, soybeans continued to flower and pods developed in August [49] with high CpRVI values.

C. Comparison of CpRVI With Full-Pol Indices

We compare three vegetation indices (i.e., RVI, GRVI, and CpRVI) for different phenological stages of wheat and soybean, as shown in Fig. 9. From several sampling sites, three representative fields (field numbers: 220, 105, 233 for wheat, and 113, 82, 232 for soybean) for each crop are used for the temporal analysis of vegetation indices. The temporal trend follows PAI and VWC for both wheat and soybean.

Similar to the analysis reported in Section IV-A for wheat, the field measurements are compared among the three groups that are separated based on PD (PD: number of plants per square meter). The growth trends of the vegetation indices are similar irrespective of PD among several fields, i.e., an increase of PAI and VWC with crop development. However, the RVI, GRVI, and CpRVI values for field no. 105 are comparatively higher for all phenological observations that have the highest PD (approximately 230 plants/m^2). In the

TABLE III
CORRELATION OF VEGETATION INDICES WITH PAI AND VWC FOR WHEAT AND SOYBEAN

Crop	Vegetation parameter	Correlation coefficient (r)		
		CpRVI	GRVI	RVI
Wheat	PAI	0.72	0.77	0.68
	VWC	0.62	0.71	0.60
Soybean	PAI	0.85	0.92	0.76
	VWC	0.75	0.82	0.74

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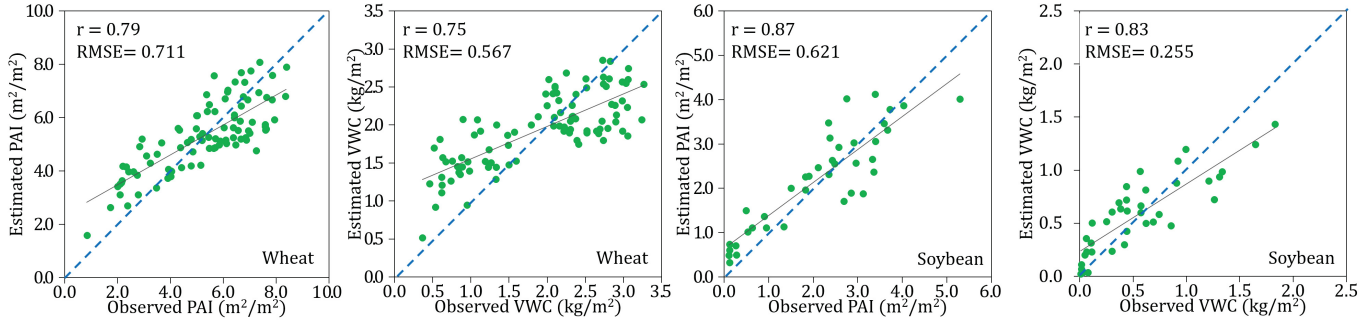


Fig. 10. Validation plots of PAI ($m^2\,m^{-2}$) and VWC ($kg\,m^{-2}$) retrieval for wheat and soybean.

other two fields, separation of CpRVI, GRVI, and RVI values within a field becomes apparent (see Fig. 9) when wheat has advanced from the leaf development to booting and start of the flowering stage.

CpRVI values reached their maximum when crop advanced from stem elongation to early dough or fruit development stages on 17 July. CpRVI values reach up to 0.65 for low PD fields, while these values peak at ≈0.8 for high PD wheat fields. This variation may be due to the high degree of randomness in scattering from the canopy elements during the flowering to fruit development stages. Unlike CpRVI, which follows almost a monotonic increase along phenological stages, the fluctuation of RVI values is more apparent. It is possibly due to the inherent noise that arises in the numerator term of RVI formulation, i.e., the third eigenvalue, which is more affected by noise rather than changes in vegetation randomness. However, the GRVI values are more stable than the RVI and CpRVI.

Unlike the RVI, which models the vegetation layer as an aggregation of randomly oriented dipoles, the GRVI offers the flexibility to choose the parameters of the GVSM from the measured data to describe the volume scattering model [42]. In this way, the GVSM might capture the phenological changes within a volume model, whereas both CpRVI and RVI lack this aspect. On the contrary, in the absence of any volume scattering models to suitably describe scattering from vegetation in CP-SAR data, the ID is utilized to measure randomness from distributed targets in CpRVI. Moreover, GDs from the odd (i.e., trihedral) and even bounce (i.e., dihedral) scatterers that appear in GRVI [42] are suitably replaced by the received echo powers from the SC and the OC sense to the transmitted polarization.

The variations of CpRVI and RVI values are more apparent in the low-biomass soybean crop. Fig. 9 shows the temporal trends of CpRVI and RVI for three representative soybean fields with different RS and PC per meter length. Both the CpRVI and full-pol indices values for each field increase as the vegetation growth increases from the early vegetative growth stages to the beginning of pod development. With the increase in vegetation components, the differential increase in vegetation indices values among several fields is apparent. The correlation analysis of vegetation indices with crop biophysical parameters also indicates a higher correlation (r) of CpRVI than RVI, as given in Table III. The correlation coefficient (r) of CpRVI with PAI is 0.72 and 0.85, which is higher than that of RVI ($r = 0.68$ and 0.76) for wheat and soybean. A similar improvement of r is observed for VWC for both the crops. The correlation analysis shows marginally better performance of CpRVI compared with RVI, while it is inferior to GRVI for characterizing vegetation growth.

D. PAI and VWC Estimation

We fit linear regression models for explaining the biophysical parameters PAI and VWC using CpRVI for several sampling locations (see Fig. 4 and Table IV). The retrieval results are presented in Fig. 10 with a 1:1 plot for wheat and soybean. We measure the accuracy of PAI and VWC retrieval by the correlation coefficient (r) and rms error.

For wheat, the estimated PAI closely follows the 1:1 line with $r = 0.79$ and RMSE $= 0.711\ m^2\ m^{-2}$. Higher dispersion of estimates is observed throughout the entire range of VWC with $r = 0.75$ and RMSE $= 0.567\ kg\,m^{-2}$. The error may have propagated from the CpRVI-VWC linear regression model (see Fig. 4). However, for both PAI and VWC retrieval, the overestimations are apparent in the early development stage of wheat. At this early stage, backscatter response is dominated by soil moisture contributions due to low leaf area [62] as reported for PAI and biomass estimation for wheat using this data set [42], [63]. Moreover, the assumption of random volume scattering from wheat fields is quite challenging to satisfy. It is because wheat has erectophile leaf orientation, satisfying a predominant vertical structure instead of a random volume structure. However, volume scattering can occur due to multiple interactions of the EM wave with stems, leaves, and the underlying soil [64] when the canopy density is high.

On the other hand, the structure of the soybean canopy is random with secondary and tertiary stems and randomly oriented leaves. During the full vegetative stage, the soybean canopy forms a very lossy volume medium, as indicated in [57]. Hence, the observed $\mathbf{K}$ matrix might be similar to $\mathbf{K}_{\mathrm{ID}}$ under the GD framework. The estimated PAI follows the 1:1 line, with $r = 0.87$ and RMSE $= 0.621\ m^2\,m^{-2}$, while the VWC estimation shows $r = 0.83$ and RMSE $= 0.255\ kg\,m^{-2}$. It is evident from Fig. 10 that the dispersion of the estimates throughout the entire range of PAI and VWC is lower for soybean compared with wheat. Nonetheless, errors might be propagating from the CpRVI-PAI and CpRVI-VWC models

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TABLE IV
REGRESSION MODEL COEFFICIENTS AND STATISTICS OF CPRVI VERSUS PAI ($m^2\,m^{-2}$) AND VWC ($kg\,m^{-2}$) FOR WHEAT AND SOYBEAN

Crop	Parameter	Regression model	r	F-stat	p-value
Wheat	PAI	$\text{PAI} = -2.062 + 12.416 \cdot \text{CpRVI}$	0.72	27.68	0.0034
	VWC	$\text{VWC} = -0.904 + 4.821 \cdot \text{CpRVI}$	0.62	23.44	0.0087
Soybean	PAI	$\text{PAI} = -1.035 + 6.483 \cdot \text{CpRVI}$	0.85	44.67	0.0006
	VWC	$\text{VWC} = -0.392 + 1.971 \cdot \text{CpRVI}$	0.75	36.75	0.0021

(see Fig. 4) into the linear regression. In both PAI and VWC retrievals, overestimation is apparent in the early development stages that are likely due to the underlying soil contribution to total backscatter power [4].

V. CONCLUSION

We have proposed a novel RVI (CpRVI) for CP-SAR data using the concept of a GD between the observed Kennaugh matrix and the Kennaugh matrix for an ID. The temporal analysis of the proposed CpRVI derived from the simulated RCM data suggests that it follows crop growth development, i.e., increasing with an increase in PAI and VWC. Also, a positive correlation with these crop biophysical parameters confirms the potential of CpRVI.

CpRVI followed the advancement of plant growth until full canopy development with the accumulation of PAI and biomass, which is evident from the high correlation with these parameters. However, with the maturity of crops, the reduction of CpRVI is not apparent. We hypothesize that CpRVI is related to the similarity between the observed Kennaugh matrix **K** and that of the ID, which depends on both the plant dielectric constant (water content) and structure of the canopy. Thus, the decrease in CpRVI values would be evident for late senescence (before harvest) when the plant VWC reduces to a minimum. Both PAI and VWC were accurately estimated using an empirical model with CpRVI.

Notably, the proposed CpRVI for CP-SAR data holds significant interest from an operational perspective for upcoming SAR missions, e.g., the RCM, RISAT-1A, and NISAR; all of them will be collecting data in the CP mode. In particular, the RCM will be capable of acquiring CP data at swaths of up to 500 km with different NESZ values. Hence, monitoring of crop growth over national scales with compact-pol indices would be a good proxy. However, implications with real compact-pol data in terms of transmitted circularity, NESZ, and hardware limitations need to be further explored, as this may affect the scattering response from targets. Moreover, experimental validation of vegetation indices on the incidence angle variations is essential for wide swath RCM products. Nevertheless, the effectiveness of the proposed CpRVI for large swath observation needs further studies with experiment setups (*in situ* measurements and satellite passes) to provide a conclusive exposition.

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Dipankar Mandal (Student Member, IEEE) received the B.Tech. degree in agricultural engineering from Bidhan Chandra Krishi Viswavidyalaya, Mohanpur, India, in 2015. He is currently pursuing the joint M.Tech./Ph.D. degree in geoinformatics and natural resources engineering with the Centre of Studies in Resources Engineering, IIT Bombay, Mumbai, India.

He was a Visiting Researcher with the Agriculture and Agri-Food Canada (AAFC), Ottawa, ON, Canada, and Carleton University, Ottawa, from October 2018 to February 2019. As a Visiting Researcher, he contributed to synthetic aperture radar (SAR) intercomparison experiment for crop biophysical parameter estimation within the Joint Experiment for Crop Assessment and Monitoring (JECAM) network of GEO Global Agricultural Monitoring. His research interests include applications of SAR polarimetry for crop classification, vegetation biophysical parameter estimation, deriving vegetation indices, and yield forecasting.

Mr. Mandal was a recipient of the Shastri Research Student Fellowship Award by the Shastri Indo-Canadian Institute, India, from 2018 to 2019.



Debanshu Ratha (Student Member, IEEE) received the five-year integrated M.Sc. degree in mathematics from the National Institute of Science Education and Research (NISER), Bhubaneswar, Odisha, in 2013. He is currently pursuing the Ph.D. degree with the Centre of Studies in Resources Engineering (CSRE), IIT Bombay, Mumbai, India.

Since then, he worked in full-time research positions at Delft Institute of Applied Mathematics (DIAM), Delft University of Technology (TU Delft), Delft, The Netherlands, and CSRE, IIT Bombay. He has been a Visiting Researcher with the Institut d'Électronique et de Télécommunications de Rennes (IETR) Laboratory, University of Rennes 1, Rennes, France, through the Raman Charpak Fellowship Program. His research interests include the theory of remote sensing and its applications, radar polarimetry, variational analysis, information geometry, graph theory, machine learning, and data fusion.

Mr. Ratha was a recipient of many prestigious scholarships and visiting fellowships in the past. He received the CSIR Fellowship Award by the CSRE, IIT Bombay.



Avik Bhattacharya (Senior Member, IEEE) received the integrated M.Sc. degree in mathematics from IIT Kharagpur, Kharagpur, India, in 2000, and the Ph.D. degree in remote sensing image processing and analysis from Télécom ParisTech, Paris, France, and the Ariana Research Group, Institut National de Recherche en Informatique et en Automatique (INRIA), Nice, France, in 2007.

He was a Canadian Government Research Fellow with the Canadian Centre for Remote Sensing (CCRS), Ottawa, ON, Canada. He is currently an Associate Professor with the Centre of Studies in Resources Engineering, IIT Bombay (CSRE, IITB), Mumbai, India, where he is also leading the Microwave Remote Sensing Laboratory. His research interests include synthetic aperture radar (SAR) polarimetry, statistical analysis of polarimetric SAR images, and applications of radar remote sensing in agriculture, cryosphere, urban, and planetary studies.

Dr. Bhattacharya received the Natural Sciences and Engineering Research Council of Canada Visiting Scientist Fellowship at the Canadian national laboratories, from 2008 to 2011. He is the Founding Chairperson of the IEEE Geoscience and Remote Sensing Society (GRSS) Chapter of the Bombay Section. He was an Associate Editor of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). He has been a Guest Editor of the Special Issue on Applied Earth Observations and Remote Sensing in India in the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (J-STARS) in 2017. He was one of the guest editors of the special stream on Advanced Statistical Techniques in SAR Image Processing and Analysis in the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2018. He is also the Editor-in-Chief of the IEEE GRSL.



Vineet Kumar (Member, IEEE) received the B.Tech. degree in electronics and communication engineering from Uttar Pradesh Technical University, Lucknow, India, in 2010, the M.Tech. degree in remote sensing and geographic information system (GIS) from the Indian Institute of Remote Sensing, Dehradun, India, in 2013, and the Ph.D. degree in synthetic aperture radar (SAR) remote sensing from IIT Bombay, Mumbai, India, in 2019.

He was a Visiting Researcher with Carleton University, Ottawa, ON, Canada, and the Agriculture and Agri-Food Canada (AAFC), Ottawa, from April to July 2016. His research interests include applications of full and compact SAR polarimetry, for classification and characterization of vegetation, and relating biophysical variables of crops to SAR observables.

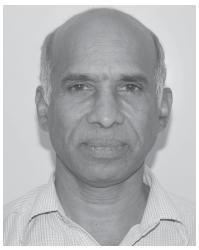
Dr. Kumar was a recipient of the Shastri Research Student Fellowship 2016 Award by the Shastri Indo-Canadian Research Institute, India.



Heather McNairn received the bachelor's degree in environmental studies from the University of Waterloo, Waterloo, ON, Canada, in 1987, the master's degree in soil science from the University of Guelph, Guelph, ON, Canada, in 1991, and the Ph.D. degree in geography from Université Laval, Quebec, QC, Canada, in 1999.

She is currently a Senior Scientist with the Agriculture and Agri-Food Canada, Ottawa, ON, Canada. She has 25 years of experience researching methods to monitor crops and soil using multispectral, hyperspectral, and synthetic aperture radar sensors. She is also an Adjunct Professor with the University of Manitoba, Winnipeg, MB, Canada, and Carleton University, Ottawa.

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Yalamanchili S. Rao (Member, IEEE) received the M.Sc. degree in physics from Andhra University, Visakhapatnam, India, in 1982, and the Ph.D. degree in passive microwave remote sensing of soil moisture from IIT Bombay, Mumbai, India, in 1992.

From 2005 to 2009, he was a Senior Research Scientist and then Associate Professor from 2009 to 2014. He was engaged in both passive and active microwave remote sensing for several applications viz., soil moisture, crop monitoring, flood mapping, and land use/land cover. He joined the Centre of Studies Resources Engineering, IIT Bombay, in 1985, as a Senior Research Assistant, and then became a Research Scientist in 1999. He is currently a Professor with IIT Bombay. He has participated in several spaceborne campaigns for collecting synchronous ground-truth data and has experience in handling various data sets for several applications. His research interests include the application of polarimetry for bio-geophysical parameter retrieval and synthetic aperture radar (SAR) interferometry for digital elevation model (DEM) and displacement map generation.



Alejandro C. Frery (Senior Member, IEEE) received the B.Sc. degree in electronic and electrical engineering from Universidad de Mendoza, Mendoza, Argentina, in 1985, the M.Sc. degree in applied mathematics (Statistics) from the Instituto Nacional de Matemática Pura e Aplicada (IMPA), Rio de Janeiro, Brazil, in 1990, and the Ph.D. degree in applied computing from the Instituto Nacional de Pesquisas Espaciais (INPE), São José dos Campos, Brazil, in 1993.

He is currently the Leader of the Laboratório de Computação Científica e Análise Numérica (LaCCAN), Universidade Federal de Alagoas, Maceió, Brazil, and also holds a Huashan Scholar position at the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, Xi'an, China, for the period of 2019–2021. His research interests include statistical computing and stochastic modeling.

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